

In the case of latent diffusion training, the encoder is utilized to obtain the latent representations of the images during the forward diffusion process, which progressively adds noise at each step. Conversely, during inference, the denoised latents generated by the reverse diffusion process are converted back into images using only the VAE decoder. Hence, during inference, only the VAE decoder is necessary.



What is the U-Net?

Introduced in 2015, U-Nets were originally proposed as a solution to the problem of biomedical image segmentation. What sets them apart is their ability to take an image as input and output the amount of noise added to each pixel. They have a unique structure where the input and output shapes are the same. U-Nets are made up of residual layers, downsampling convolutions, and upsampling convolutions, with skip connections connecting layers with the same spatial size in the two halves of the network. The downsampling convolutions capture the general context of the image, while the skip connections provide detailed information to the upsampling layers.

The U-Net consists of two parts, an encoder and a decoder, each composed of ResNet blocks. The encoder compresses an image representation into a lower-resolution image representation, and the decoder decodes the lower-resolution image representation back to the original higher-resolution image representation, which is expected to be less noisy. The U-Net output predicts the noise residual, which can be used to compute the denoised image representation. To prevent the U-Net from discarding important information while downsampling, shortcut connections are added between